THE SOCIAL SCIENCE STATISTICAL TESTS REFERENCE GUIDE

A Decision-Tree Approach to Choosing the Right Analysis

For Graduate Students and Early-Career Researchers

Comprehensive coverage of 35+ statistical methods with examples, R code, and APA templates

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How to Use This Guide

Welcome! This guide is designed to be your go-to reference for choosing and understanding the statistical tests you will encounter most often in social science research. Whether you are working on a thesis, analyzing survey data, or reviewing a journal article, this document will help you navigate the sometimes-overwhelming world of statistical methods.

The guide is organized into three parts:

- Part 1: The Decision-Tree Flow — Start here. Walk through a series of questions about your data and research question to arrive at the right test.
- Part 2: Categorical Reference — Tests grouped by family (t-tests, ANOVA, regression, etc.) for quick lookup when you already know what you need.
- Part 3: Supplementary Material — Assumption checking, effect size tables, troubleshooting, power analysis, and a glossary.

Every test entry follows the same template: a plain-language explanation, a concrete social science example, an intuitive analogy, the formula with defined symbols, R code, and an APA reporting template. We lead with understanding before formulas — because knowing what a test does matters more than memorizing its equation.

> **Pro Tip:** If you know your test but need a quick refresher, jump to Part 2. If you are unsure which test to use, start with Part 1 and follow the decision branches.

PART 1: Decision-Tree Flow

This section walks you through a structured decision process. Start with the first question — what type of research question are you asking? — then follow the branches based on your variables, measurement levels, and sample characteristics. At each terminal node, you will find a detailed entry for the recommended statistical test.

Quick Decision Overview

Research Question	Key Follow-Up	Common Tests
Comparing groups	How many groups? Independent or paired?	t-tests, ANOVA, Mann-Whitney, Kruskal-Wallis
Examining relationships	Variable types? How many?	Pearson r, Spearman rho, Chi-square, Point-biserial
Predicting outcomes	DV type? How many predictors?	Linear regression, Logistic regression, Poisson
Examining structure	Exploring or confirming?	EFA, CFA, PCA, Cluster analysis
Change over time	Same subjects? How many timepoints?	Paired t-test, RM-ANOVA, Mixed models, Friedman

Branch 1: Comparing Groups (Differences)

You want to know whether groups differ on some outcome. The key follow-up questions are: How many groups are you comparing? Are they independent (different people in each group) or related (same people measured more than once)? Is your dependent variable continuous or categorical? And are parametric assumptions met?

Groups	Independent?	DV Type	Assumptions Met?	Test
2	Yes	Continuous	Yes	Independent-Samples t-test
2	Yes	Continuous	No	Mann-Whitney U
2	No (paired)	Continuous	Yes	Paired-Samples t-test
2	No (paired)	Continuous	No	Wilcoxon Signed-Rank
1 (vs. value)	—	Continuous	Yes	One-Sample t-test
3+	Yes	Continuous	Yes	One-Way ANOVA
3+	Yes	Continuous	No	Kruskal-Wallis H
3+	No (paired)	Continuous	Yes	Repeated-Measures ANOVA
3+	No (paired)	Continuous	No	Friedman Test
2+ (per IV)	Yes	Continuous	Yes	Factorial ANOVA
2+ (per IV)	Mixed (between + within)	Continuous	Yes	Mixed ANOVA
2+	Yes	Continuous (with covariate)	Yes	ANCOVA
2+	Yes	Multiple DVs	Yes	MANOVA

One-Sample t-Test

Also known as: *Single-sample t-test*

What It Does

The one-sample t-test determines whether the mean of a single sample differs significantly from a known or hypothesized population value. It asks: "Is my sample's average meaningfully different from this specific number, or could the difference just be due to random sampling variability?"

Concrete Example

Imagine you are studying whether psychology graduate students sleep a different amount than the national average of 7 hours per night. You measure sleep duration in 40 grad students and get a mean of 6.2 hours. The one-sample t-test tells you whether that 0.8-hour difference is statistically significant or could easily be explained by chance.

Intuitive Analogy

Think of it like weighing a bag of coffee that says "12 oz" on the label. You weigh it once, get 11.6 oz, and wonder: is the bag actually short, or is this just normal variation in filling? The one-sample t-test is your formal way of making that judgment — comparing one measurement (your sample mean) against a claimed standard (the label).

When and Why to Use It

Use this test when you have one continuous variable measured on one group and you want to compare that group's mean against a specific reference value (a population mean, a benchmark, a theoretical value). It is the simplest parametric comparison test.

Key Assumptions

- The dependent variable is continuous (interval or ratio scale).
- The observations are independent of each other (one person's score does not influence another's).
- The data are approximately normally distributed (especially important for small samples, n < 30). For larger samples, the Central Limit Theorem helps.
- No significant outliers that could distort the mean.

Formula

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

$\bar{x}$ = Sample mean

μ_0 = Hypothesized population mean (your comparison value)

s = Sample standard deviation

n = Sample size

$s / \sqrt{n}$ = Standard error of the mean

Effect Size

Cohen's d = $(\bar{x} - \mu_0) / s$. Interpretation: d = 0.2 (small), d = 0.5 (medium), d = 0.8 (large). This tells you how many standard deviations your sample mean is from the hypothesized value.

R Implementation

Package: Base R (no additional packages needed)

```
# One-sample t-test: comparing sample mean to mu = 7
sleep_hours <- c(6.5, 5.8, 6.1, 7.0, 6.3, 5.9, 6.8, 6.0)
t.test(sleep_hours, mu = 7)

# Effect size with effectsize package
# install.packages("effectsize")
library(effectsize)
cohens_d(sleep_hours, mu = 7)
```

APA Reporting Template

APA Reporting: *A one-sample t-test indicated that graduate students' mean sleep duration (M = 6.20, SD = 1.44) was significantly lower than the national average of 7 hours, t(39) = -3.52, p = .001, d = 0.56.*

What If Assumptions Are Violated?

If normality is violated with a small sample, consider the Wilcoxon signed-rank test as a non-parametric alternative. For large samples (n > 30), the t-test is generally robust to mild non-normality due to the Central Limit Theorem.

Independent-Samples t-Test

Also known as: *Two-sample t-test, Student's t-test, Welch's t-test*

What It Does

This test compares the means of two independent (unrelated) groups to determine if they differ significantly. The groups contain different participants — what happens in one group is completely separate from the other. It asks: "Are these two groups genuinely different on this outcome, or is the gap just noise?"

Concrete Example

You randomly assign 60 participants to either a cognitive-behavioral therapy (CBT) group or a waitlist control group, then measure depression scores after 8 weeks. The independent-samples t-test tells you whether the CBT group's mean depression score is significantly lower than the control group's.

Intuitive Analogy

Imagine two different classrooms take the same exam. You want to know if one class performed better than the other — not just by a point or two that could be random, but meaningfully better. The independent t-test is your formal comparison, accounting for how spread out the scores are within each class.

When and Why to Use It

Use when comparing means of exactly two independent groups on a continuous dependent variable. If you have more than two groups, use ANOVA instead. The Welch version (default in R) is recommended because it does not assume equal variances.

Key Assumptions

- Dependent variable is continuous.
- The two groups are independent (no participant appears in both).
- Data in each group are approximately normally distributed (or n is large enough for CLT).
- Homogeneity of variance: both groups have roughly similar spread. Welch's t-test relaxes this assumption.
- No significant outliers.

Formula

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s^2_1/n_1 + s^2_2/n_2)} \quad \text{[Welch's version]}$$

$\bar{x}_1, \bar{x}_2$ = Means of groups 1 and 2

s^2_1, s^2_2 = Variances of groups 1 and 2

n_1, n_2 = Sample sizes of groups 1 and 2

Effect Size

Cohen's d = $(\bar{x}_1 - \bar{x}_2)$ / s_pooled. Benchmarks: 0.2 (small), 0.5 (medium), 0.8 (large).

R Implementation

Package: Base R (Welch's is default)

```
# Independent-samples t-test (Welch's by default)
cbt_scores <- c(12, 18, 14, 20, 15, 13, 19, 17)
control_scores <- c(16, 22, 18, 24, 19, 17, 23, 21)
t.test(cbt_scores, control_scores)

# For Student's (equal variances assumed):
t.test(cbt_scores, control_scores, var.equal = TRUE)
```

APA Reporting Template

APA Reporting: *An independent-samples t-test revealed that the CBT group (M = 16.00, SD = 2.93) reported significantly lower depression scores than the control group (M = 20.00, SD = 2.93), t(14) = -2.73, p = .016, d = 1.37.*

What If Assumptions Are Violated?

If normality is violated, use the Mann-Whitney U test. If variances are unequal, use Welch's t-test (the default in R). For very small samples, bootstrapped t-tests are an option.

Paired-Samples t-Test

Also known as: *Dependent-samples t-test, matched-pairs t-test, within-subjects t-test*

What It Does

This test compares two means from the same group of participants measured at two different times or under two different conditions. It works with the difference scores — for each person, it calculates pre minus post (or condition A minus condition B) and tests whether the average difference is significantly different from zero.

Concrete Example

You measure anxiety levels in 30 participants before and after an 8-week mindfulness program. Each person serves as their own control. The paired t-test examines whether the average change in anxiety scores is significantly different from zero.

Intuitive Analogy

It is like a before-and-after weigh-in at a fitness program. Instead of comparing two random groups of people, you are looking at how each individual changed. Did enough people lose enough weight for us to conclude the program works, beyond what random fluctuation would produce?

When and Why to Use It

Use when you have two measurements on the same participants (pre-post designs, crossover studies, matched pairs). This design is powerful because it controls for individual differences — each person serves as their own baseline.

Key Assumptions

- The difference scores (post minus pre for each participant) are continuous.
- Observations are independent (one participant's change does not affect another's).
- The difference scores are approximately normally distributed.
- No extreme outliers in the difference scores.

Formula

$$t = \bar{d} / (s_d / \sqrt{n})$$

$\bar{d}$ = Mean of the difference scores

s_d = Standard deviation of the difference scores

n = Number of pairs

Effect Size

Cohen's d_z = $\bar{d}$ / s_d (using the SD of differences). Or use d_av = $\bar{d}$ / average SD across timepoints. Benchmarks: 0.2, 0.5, 0.8.

R Implementation

Package: Base R

```
# Paired-samples t-test
pre <- c(45, 52, 38, 60, 55, 47, 50, 43)
post <- c(38, 48, 30, 46, 52, 38, 36, 44)
t.test(pre, post, paired = TRUE)

# Effect size
library(effectsize)
cohens_d(pre, post, paired = TRUE)
```

APA Reporting Template

APA Reporting: *A paired-samples t-test indicated that anxiety scores decreased significantly from pre-intervention (M = 48.75, SD = 7.01) to post-intervention (M = 41.50, SD = 7.23), t(7) = 3.92, p = .006, d = 1.39.*

What If Assumptions Are Violated?

If normality of difference scores is violated, use the Wilcoxon signed-rank test instead.

Mann-Whitney U Test

Also known as: *Wilcoxon rank-sum test, Mann-Whitney-Wilcoxon test*

What It Does

This is the non-parametric alternative to the independent-samples t-test. Instead of comparing means, it ranks all observations from both groups together and tests whether one group tends to have higher ranks than the other. It is asking: "If I pick a random score from Group A and a random score from Group B, is one systematically higher?"

Concrete Example

You compare life satisfaction scores (measured on a 1–7 Likert scale) between urban and rural residents. Because Likert data are ordinal and likely not normally distributed, you use the Mann-Whitney U instead of a t-test.

Intuitive Analogy

Imagine lining up all participants from both groups in order from lowest to highest score, then checking whether one group's members cluster toward the high end and the other toward the low end. If the groups are truly different, their members will not be randomly mixed in the lineup.

When and Why to Use It

Use when comparing two independent groups but the data violate normality, are ordinal, have significant outliers, or the sample is very small. It is always a safe alternative to the independent t-test.

Key Assumptions

- The dependent variable is at least ordinal.
- The two groups are independent.
- The distributions of the two groups have similar shapes (if you want to interpret the test as comparing medians).

Formula

$$U = n_1 n_2 + n_1(n_1+1)/2 - R_1$$

U = The Mann-Whitney U statistic

n_1, n_2 = Sample sizes of the two groups

R_1 = Sum of ranks for group 1

Effect Size

r = Z / √N (where N = total sample size). Benchmarks: 0.1 (small), 0.3 (medium), 0.5 (large). Also commonly reported: rank-biserial correlation.

R Implementation

Package: Base R

```
# Mann-Whitney U test
urban <- c(5, 6, 4, 7, 5, 6, 3, 5)
rural <- c(3, 4, 2, 5, 3, 4, 2, 3)
wilcox.test(urban, rural)

# Effect size: rank-biserial correlation
library(effectsize)
rank_biserial(urban, rural)
```

APA Reporting Template

APA Reporting: *A Mann-Whitney U test indicated that urban residents reported significantly higher life satisfaction (Mdn = 5.00) than rural residents (Mdn = 3.00), U = 54.5, p = .004, r_rb = .62.*

What If Assumptions Are Violated?

This test has minimal assumptions. If the shape assumption is violated, interpret results as comparing distributions rather than medians.

Wilcoxon Signed-Rank Test

Also known as: *Wilcoxon matched-pairs signed-rank test*

What It Does

This is the non-parametric alternative to the paired-samples t-test. It ranks the absolute values of the difference scores, then considers whether positive or negative differences dominate. It asks: "Do more participants improve than worsen, and by larger amounts?"

Concrete Example

You measure patient-reported pain scores before and after acupuncture treatment. Because pain ratings on a 0–10 scale are ordinal, you use the Wilcoxon signed-rank test to determine if pain significantly decreased.

Intuitive Analogy

Imagine each person holds up a sign showing their change: green for improvement, red for worsening, and the sign's size shows how much they changed. This test checks whether the green signs are both more numerous and larger than the red ones.

When and Why to Use It

Use instead of a paired t-test when the difference scores are not normally distributed, when you have ordinal data, or when outliers are present.

Key Assumptions

- Data are paired (same participants measured twice).
- The difference scores are at least ordinal.
- The distribution of differences is approximately symmetric (for interpreting median differences).

Formula

$$W = min(W+, W-)$$

W+ = Sum of ranks for positive differences

W− = Sum of ranks for negative differences

W = The smaller of the two rank sums

Note: **R's `wilcox.test()` reports the test statistic as V (the sum of positive ranks, equivalent to W+), not as min(W+, W−). The APA template below uses V to match R output. If computing by hand using the W = min(W+, W−) formula, your value will differ from R's but both approaches yield the same p-value.**

Effect Size

r = Z / √n (where n = number of non-zero differences). Same benchmarks: 0.1, 0.3, 0.5.

R Implementation

Package: Base R

```
# Wilcoxon signed-rank test
pre_pain  <- c(7, 8, 6, 9, 5, 7, 8, 6)
post_pain <- c(4, 5, 4, 6, 3, 5, 5, 4)
wilcox.test(pre_pain, post_pain, paired = TRUE)


# Effect size: rank-biserial correlation
library(effectsize)
rank_biserial(pre_pain, post_pain, paired = TRUE)
```

APA Reporting Template

APA Reporting: *A Wilcoxon signed-rank test indicated that pain scores were significantly lower after acupuncture treatment (Mdn = 4.50) than before (Mdn = 7.00), V = 36, p = .008, r = .65.*

What If Assumptions Are Violated?

Very few assumptions to violate. If you have very small samples (n < 6), exact p-values may be limited.

One-Way ANOVA

Also known as: *One-way analysis of variance, single-factor ANOVA*

What It Does

One-way ANOVA tests whether the means of three or more independent groups differ significantly. Rather than running multiple t-tests (which inflates Type I error), ANOVA performs a single omnibus test. It compares the variance between groups to the variance within groups. If the between-group variance is much larger than the within-group variance, the groups probably differ.

Concrete Example

You want to compare job satisfaction across three departments: marketing, engineering, and HR. Each department has 25 employees who complete a satisfaction survey. ANOVA tests whether at least one department's average satisfaction differs from the others.

Intuitive Analogy

Imagine three groups of students take an exam. ANOVA asks: Is the spread of the group averages larger than you would expect given how much scores vary within each group? If Group A averages 85, Group B 72, and Group C 89, but individual scores within each group range from 50 to 100, maybe those group differences are not that impressive. ANOVA formalizes this reasoning.

When and Why to Use It

Use when comparing means across three or more independent groups with one independent variable (the grouping factor) and one continuous dependent variable. If you only have two groups, a t-test is sufficient (and mathematically equivalent to ANOVA with 2 groups).

Key Assumptions

- The dependent variable is continuous.
- The groups are independent (different participants in each group).
- The DV is approximately normally distributed within each group.
- Homogeneity of variance: the spread of scores is similar across groups (test with Levene's test).
- No significant outliers.

Formula

$$F = MS_between \,/\, MS_within$$

$\boldsymbol{F}$ = The F-ratio (test statistic)

MS_between = Mean square between groups = SS_between / df_between

MS_within = Mean square within groups = SS_within / df_within

SS = Sum of squares (measure of variability)

df_between = k - 1, where k = number of groups

df_within = N - k, where N = total sample size

Effect Size

Eta-squared (η^2) = SS_between / SS_total. Partial eta-squared for factorial designs. Omega-squared (ω^2) is a less biased alternative. Benchmarks: 0.01 (small), 0.06 (medium), 0.14 (large).

R Implementation

Package: Base R; car package for Type III SS; effectsize package for effect sizes

```
# One-Way ANOVA
satisfaction <- c(72,68,75,80,71, 65,60,58,70,63, 85,82,90,78,88)
department <- factor(rep(c("Marketing","Engineering","HR"), each=5))
model <- aov(satisfaction ~ department)
summary(model)

# Post-hoc pairwise comparisons
TukeyHSD(model)

# Effect size
library(effectsize)
eta_squared(model)
```

APA Reporting Template

APA Reporting: *A one-way ANOVA revealed a significant effect of department on job satisfaction, $F(2, 12) = 14.32$, $p < .001$, $\eta^2 = .70$. Tukey post-hoc tests indicated that HR ($M = 84.60$) scored significantly higher than Engineering ($M = 63.20$), $p < .001$.*

What If Assumptions Are Violated?

If homogeneity of variance is violated, use Welch's ANOVA: oneway.test(dv ~ group, data = df). If normality is violated, use the Kruskal-Wallis test. ANOVA is fairly robust to mild normality violations with balanced group sizes.

Kruskal-Wallis H Test

Also known as: *Kruskal-Wallis one-way ANOVA by ranks*

What It Does

The non-parametric alternative to one-way ANOVA. It ranks all observations together, then tests whether the average ranks differ across three or more groups. It does not require normality or equal variances.

Concrete Example

You compare attitudes toward immigration (measured on a Likert scale) across four political affiliations. Since the data are ordinal and skewed, you use Kruskal-Wallis instead of ANOVA.

Intuitive Analogy

Same lineup analogy as Mann-Whitney, but now with three or more groups. Line everyone up from lowest to highest score and check whether members of each group cluster in different parts of the line.

When and Why to Use It

Use when you have three or more independent groups, an ordinal or non-normal continuous DV, or when ANOVA assumptions are seriously violated.

Key Assumptions

- The dependent variable is at least ordinal.
- Groups are independent.
- Distributions have similar shapes across groups (for median interpretation).

Formula

$$H = [12 / N(N+1)] \sum (R^2_i / n_i) - 3(N+1)$$

H = The Kruskal-Wallis test statistic (approximately chi-squared distributed)

N = Total sample size

R_i = Sum of ranks in group i

n_i = Sample size of group i

k = Number of groups

Effect Size

Epsilon-squared (ε^2) = H / ((N² - 1)/(N + 1)). Or eta-squared based on H. Benchmarks similar to eta-squared: 0.01, 0.06, 0.14.

R Implementation

Package: Base R; FSA or PMCMRplus for post-hoc Dunn tests

```
# Kruskal-Wallis test
scores <- c(5,6,4,7,5, 3,4,2,5,3, 6,7,5,6,7, 4,3,5,4,3)
group <- factor(rep(c("A","B","C","D"), each = 5))
kruskal.test(scores ~ group)

# Post-hoc: Dunn's test
library(FSA)
dunnTest(scores ~ group, method = "bonferroni")
```

APA Reporting Template

APA Reporting: *A Kruskal-Wallis test revealed a significant difference in attitudes across political affiliations, H(3) = 12.45, p = .006, ε² = .15. Post-hoc Dunn tests with Bonferroni correction indicated significant differences between Groups A and B (p = .003).*

What If Assumptions Are Violated?

Minimal assumptions. If you have tied ranks, most software handles these automatically.

Repeated-Measures ANOVA

Also known as: *Within-subjects ANOVA*

What It Does

Tests whether the mean of a continuous variable changes significantly across three or more time points or conditions for the same participants. It is the extension of the paired t-test to more than two conditions.

Concrete Example

You measure stress levels in 40 medical students at three time points: beginning of the semester, during midterms, and during finals. Repeated-measures ANOVA tests whether average stress significantly differs across these three timepoints.

Intuitive Analogy

Imagine tracking the same person's weight every month for six months during a diet program. Instead of asking "is the start different from the end?" (that is a paired t-test), you are asking "does weight change at all over the six months?" The repeated-measures ANOVA captures the full trajectory.

When and Why to Use It

Use when the same participants are measured at three or more time points or under three or more conditions on the same continuous DV.

Key Assumptions

- Continuous DV.
- Same participants at all time points (within-subjects).
- Normality of the DV at each time point.
- Sphericity: the variances of differences between all pairs of conditions are equal (test with Mauchly's test). This is the unique and often-violated assumption of RM-ANOVA.

Formula

$$F = MS_conditions \,/\, MS_error$$

MS_conditions = Variance between conditions

MS_error (also written MS_S×C) = Residual variance after removing between-subjects and between-conditions variance; represents the subjects × conditions interaction

Effect Size

Partial eta-squared (η^2_p). Generalized eta-squared (η^2_G) is preferred for within-subjects designs. Benchmarks: 0.01, 0.06, 0.14.

R Implementation

Package: Base R for aov(); ez package for ezANOVA (handles sphericity corrections); afex for an alternative ANOVA implementation (aov_ez)

```
# Repeated-Measures ANOVA with ez
library(ez)
df <- data.frame(
  id = factor(rep(1:10, 3)),
  time = factor(rep(c("Start","Mid","Final"), each = 10)),
  stress = c(35,40,38,42,37,39,41,36,44,38,
             50,55,52,58,48,53,56,49,60,54,
             45,48,44,50,42,46,49,43,52,47))
```

```
ezANOVA(data=df, dv=stress, wid=id, within=time)
```

APA Reporting Template

APA Reporting: *A repeated-measures ANOVA revealed a significant effect of time on stress levels, $F(2, 18) = 32.15$, $p < .001$, $\eta^2_G = .52$. Greenhouse-Geisser correction was applied ($\varepsilon = .82$).*

What If Assumptions Are Violated?

If sphericity is violated (Mauchly's test $p < .05$), apply a correction: Greenhouse-Geisser (conservative) or Huynh-Feldt (liberal). If normality is violated, use the Friedman test.

Friedman Test

Also known as: *Friedman's ANOVA by ranks*

What It Does

The non-parametric alternative to repeated-measures ANOVA. It ranks observations within each participant across conditions, then tests whether the average ranks differ across conditions.

Concrete Example

Participants rate three different therapy approaches (CBT, DBT, ACT) on perceived helpfulness using a 1–10 scale. Since the ratings are ordinal, you use the Friedman test.

Intuitive Analogy

Each participant ranks the three approaches from best to worst. The Friedman test checks whether one approach consistently gets higher rankings across participants.

When and Why to Use It

Use when you have three or more related/repeated conditions, ordinal data, or when RM-ANOVA assumptions are violated.

Key Assumptions

- Data are at least ordinal.
- Same participants across all conditions.
- Observations within each participant are ranked across conditions.

Formula

$$\chi^2_r = [12 / nk(k+1)] \sum R^2_j - 3n(k+1)$$

χ^2_r = Friedman test statistic (chi-squared distributed)

n = Number of participants

k = Number of conditions

R_j = Sum of ranks for condition j

Effect Size

Kendall's W = χ^2_r / (n(k-1)). Ranges 0 to 1, where 0 indicates no concordance and 1 indicates perfect concordance. Values above .50 are often interpreted as moderate-to-strong concordance, though no universally agreed-upon benchmarks exist for Kendall's W.

R Implementation

Package: Base R; PMCMRplus for post-hoc tests

```
# Friedman test
ratings <- matrix(c(7,5,8, 6,4,9, 8,6,7, 5,3,8, 7,5,9),
                  nrow=5, byrow=TRUE)
colnames(ratings) <- c("CBT","DBT","ACT")
friedman.test(ratings)
```

APA Reporting Template

APA Reporting: *A Friedman test indicated a significant difference in perceived helpfulness across therapy types, $\chi^2(2) = 8.40$, $p = .015$, $W = .84$.*

What If Assumptions Are Violated?

Minimal assumptions. For post-hoc pairwise comparisons, use Nemenyi or Conover tests from PMCMRplus.

Factorial ANOVA

Also known as: *Two-way ANOVA, multi-factor ANOVA*

What It Does

Factorial ANOVA examines the effects of two or more independent variables (factors) on a continuous dependent variable simultaneously. Its key advantage is testing interaction effects — whether the effect of one IV depends on the level of another IV.

Concrete Example

You study how both therapy type (CBT vs. medication) and severity (mild vs. severe) affect depression outcomes. Factorial ANOVA tells you: (1) Does therapy type matter? (2) Does severity matter? (3) Crucially, does the effectiveness of therapy type depend on severity? That third question is the interaction.

Intuitive Analogy

Think of baking a cake. Changing the oven temperature affects the result, and changing the flour type affects the result. But some flour types perform best at specific temperatures — that is the interaction. Factorial ANOVA detects these conditional effects.

When and Why to Use It

Use when you have two or more categorical IVs, a continuous DV, and independent groups. The most common version is 2 x 2 (two IVs, each with two levels).

Key Assumptions

- Continuous DV.
- Independent observations.
- Normality of DV within each cell (combination of factor levels).
- Homogeneity of variance across all cells (Levene's test).
- Balanced or near-balanced designs are preferred (use Type III SS for unbalanced).

Formula

$$F_A = MS_A / MS_error;\ \ F_B = MS_B / MS_error;\ \ F_AxB = MS_AxB / MS_error$$

F_A = F-ratio for factor A main effect

F_B = F-ratio for factor B main effect

F_AxB = F-ratio for the A × B interaction

MS = Mean square (variance estimate) for each source

Effect Size

Partial eta-squared (η^2_p) for each main effect and the interaction separately. Benchmarks: 0.01, 0.06, 0.14. Omega-squared is less biased.

R Implementation

Package: Base R for aov(); car package for Type III SS with Anova()

Important: When using base R's aov() alongside car::Anova() for Type III Sums of Squares, you must set orthogonal contrasts first to get mathematically accurate p-values. Note that options(contrasts = ...) is a global, session-persistent setting — it affects all models run afterward in the same R session. Reset to defaults after your analysis with: options(contrasts = c("contr.treatment", "contr.poly")).

```
# 2x2 Factorial ANOVA
library(car)
df <- data.frame(
   score = c(45,50,42,48,55,60,52,58, 35,30,38,32,50,48,52,46),
   therapy = factor(rep(c("CBT","Med"), each=8)),
   severity = factor(rep(rep(c("Mild","Severe"), each=4), 2)))

# Set orthogonal contrasts for Type III SS
options(contrasts = c("contr.sum", "contr.poly"))
model <- aov(score ~ therapy * severity, data = df)
Anova(model, type = 3)
```

APA Reporting Template

APA Reporting: *A 2 × 2 factorial ANOVA revealed a significant main effect of therapy type, $F(1, 12) = 8.45$, $p = .013$, $\eta^2_p = .41$, a significant main effect of severity, $F(1, 12) = 22.10$, $p < .001$, $\eta^2_p = .65$, and a significant interaction, $F(1, 12) = 5.32$, $p = .040$, $\eta^2_p = .31$.*

What If Assumptions Are Violated?

For unequal cell sizes, always use Type III SS. If homogeneity of variance is violated, consider Welch-type corrections or use a linear model with robust standard errors.

Mixed ANOVA

Also known as: *Split-plot ANOVA, mixed-design ANOVA*

What It Does

Mixed ANOVA combines between-subjects and within-subjects factors. You have at least one factor where different groups of participants are compared (between) and at least one factor where the same participants are measured repeatedly (within). It tests main effects of each factor plus their interaction.

Concrete Example

You compare two treatment groups (CBT vs. control) measured at three time points (baseline, 4 weeks, 8 weeks). Treatment group is between-subjects; time is within-subjects. The interaction tells you whether the groups change differently over time — this is usually the most important result.

Intuitive Analogy

Two teams run a relay race over three legs. The mixed ANOVA asks: Do the teams differ overall? Does performance change across legs? And critically, does one team improve while the other declines?

When and Why to Use It

Use when your design has at least one between-subjects factor and at least one within-subjects factor. Very common in clinical trials with treatment vs. control groups measured over time.

Key Assumptions

- Continuous DV.
- Independence between groups (not within).
- Normality within each cell.
- Homogeneity of variance (between-subjects factor).
- Sphericity (within-subjects factor) — test with Mauchly's test.

Formula

$$\text{Multiple F-ratios: } F_between = MS_between / MS_subjects(between);\ F_within = MS_within / MS_within\times subjects;\ F_interaction = MS_interaction / MS_within\times subjects$$

F_between = F-ratio for between-subjects factor

F_within = F-ratio for within-subjects factor

F_interaction = F-ratio for the between × within interaction

Effect Size

Partial eta-squared or generalized eta-squared for each effect. Same benchmarks.

R Implementation

Package: ez package (ezANOVA); afex package (aov_ez)

```
library(ez)
df <- data.frame(
  id = factor(rep(1:12, each = 3)),
  group = factor(rep(c("CBT","Control"), each = 18)),
  time = factor(rep(c("Pre","Mid","Post"), 12)),
  score = c(50,40,30,55,42,28,48,38,25,52,41,32,
            60,35,20,58,38,22, 50,48,47,53,50,49,
            48,46,45,55,52,50, 52,49,48,51,50,49))
ezANOVA(data=df, dv=score, wid=id, within=time, between=group)
```

APA Reporting Template

APA Reporting: *A mixed ANOVA with a between-subjects factor of group (CBT vs. control) and within-subjects factor of time (pre, mid, post) revealed a significant group × time interaction, $F(2, 20) = 15.43$, $p < .001$, $\eta^2_G = .28$, indicating that the CBT group improved significantly more over time than the control group.*

What If Assumptions Are Violated?

If sphericity is violated, apply Greenhouse-Geisser correction. For unequal group sizes, consider a linear mixed-effects model (lme4 package) as a more flexible alternative.

ANCOVA

Also known as: *Analysis of Covariance*

What It Does

ANCOVA combines ANOVA with regression by comparing group means on a DV while statistically controlling for one or more continuous covariates. It "adjusts" the group means to what they would be if all groups had the same value on the covariate, reducing error variance and increasing statistical power.

Concrete Example

You compare three teaching methods on exam scores, but students vary in prior knowledge. ANCOVA lets you compare the teaching methods while statistically equalizing prior knowledge across groups. It asks: "After removing the influence of prior knowledge, do the teaching methods still differ?"

Intuitive Analogy

Imagine three runners racing on tracks of different elevation. ANCOVA adjusts their times as if they all ran on flat ground, so you can fairly compare their speed — the covariate (elevation) is removed from the comparison.

When and Why to Use It

Use when comparing group means on a DV while controlling for a continuous covariate that is related to the DV but not the IV. Common in quasi-experimental designs where random assignment was not possible.

Key Assumptions

- All ANOVA assumptions (normality, homogeneity of variance, independence).
- Linear relationship between covariate and DV.
- Homogeneity of regression slopes: the relationship between the covariate and DV is the same across groups (no interaction between covariate and IV).
- The covariate is measured without error (reliability).
- The covariate is not influenced by the IV (measured before treatment).

Formula

$$F = MS_adjusted_between \,/\, MS_adjusted_within$$

MS_adjusted = Mean squares after removing covariate variance

The adjustment = Partials out the covariate's effect using regression before testing group differences

Effect Size

Partial eta-squared on the adjusted F-test. Same benchmarks: 0.01, 0.06, 0.14.

R Implementation

Package: Base R; car for Type III SS

Important: When using base R's aov() alongside car::Anova() for Type III Sums of Squares, you must set orthogonal contrasts first to get mathematically accurate p-values. Note that options(contrasts = ...) is a global, session-persistent setting — it affects all models run afterward in the same R session. Reset to defaults after your analysis with: options(contrasts = c("contr.treatment", "contr.poly")).

```
library(car)
df <- data.frame(
  score = c(78,82,75,80,85, 68,72,65,70,74, 88,92,85,90,94),
  method = factor(rep(c("A","B","C"), each = 5)),
  prior = c(60,65,58,63,70, 62,68,55,64,66, 58,62,56,60,64))

# Set orthogonal contrasts for Type III SS
options(contrasts = c("contr.sum", "contr.poly"))

model <- aov(score ~ prior + method, data = df)
Anova(model, type = 3)
```

APA Reporting Template

APA Reporting: *An ANCOVA with prior knowledge as a covariate revealed a significant effect of teaching method on exam scores after controlling for prior knowledge, $F(2, 11) = 18.74$, $p < .001$, $\eta^2_p = .77$.*

What If Assumptions Are Violated?

If the homogeneity of regression slopes assumption is violated (covariate × group interaction is significant), ANCOVA results are misleading. Consider a moderated regression model instead.

MANOVA

Also known as: *Multivariate Analysis of Variance*

What It Does

MANOVA extends ANOVA to situations where you have two or more continuous dependent variables. Instead of running separate ANOVAs for each DV (which inflates Type I error), MANOVA tests them simultaneously, accounting for correlations between DVs.

Concrete Example

You compare three therapy conditions on both depression and anxiety scores simultaneously. MANOVA tests whether the therapy groups differ on the combination of depression and anxiety, capturing patterns that separate ANOVAs would miss.

Intuitive Analogy

Instead of judging a chef by taste alone or presentation alone, MANOVA is like a food competition where you rate the overall dish — considering taste, presentation, and texture together as a package.

When and Why to Use It

Use when you have two or more continuous, correlated DVs and one or more categorical IVs. The DVs should be conceptually related but not redundant.

Key Assumptions

- Multiple continuous DVs.

- Independent observations between groups.
- Multivariate normality (hard to test; check each DV's normality as an approximation).
- Homogeneity of variance-covariance matrices (Box's M test; note this test is very sensitive).
- No multicollinearity among DVs (correlations should not exceed ~.90).
- Sample size: need more cases in every cell than DVs.

Formula

Uses test statistics based on eigenvalues: Wilks' Λ, Pillai's trace, Hotelling's trace, Roy's largest root

Wilks' Λ = Most commonly reported; ranges from 0 to 1, lower = larger group difference

Pillai's trace = More robust to violations; recommended when assumptions are questionable

F approximation = Each multivariate statistic is converted to an approximate F for p-value calculation

Effect Size

Partial eta-squared from the Pillai's trace or Wilks' lambda. Multivariate eta-squared. Same benchmarks.

R Implementation

Package: Base R (manova function); car for detailed output

```
# MANOVA
depression <- c(25,20,22, 18,15,12, 30,28,26)
anxiety <- c(30,25,28, 20,18,15, 35,32,30)
group <- factor(rep(c("CBT","Med","Control"), each = 3))
model <- manova(cbind(depression, anxiety) ~ group)
summary(model, test = "Pillai")

# Follow-up univariate ANOVAs
summary.aov(model)
```

APA Reporting Template

APA Reporting: *A MANOVA revealed a significant multivariate effect of therapy condition on the combined depression and anxiety outcomes, Pillai's trace = .89, F(4, 12) = 4.56, p = .018, η²_p = .45.*

What If Assumptions Are Violated?

If Box's M is significant, use Pillai's trace (most robust). If multivariate normality is violated with small samples, consider separate robust ANOVAs with a Bonferroni correction.

Branch 2: Examining Relationships (Associations)

You want to know whether two or more variables are related — do they move together, and if so, how strongly and in what direction? The key follow-ups: What types of variables do you have? How many?

Variable 1	Variable 2	Data Characteristics	Test
Continuous	Continuous	Normal, linear	Pearson r
Continuous	Continuous (non-normal)	Non-normal or nonlinear	Spearman rho
Ordinal	Ordinal	Ordinal	Spearman rho or Kendall's tau (see Part 2: Correlation Family; Kendall's tau preferred for small n or heavy ties)
Dichotomous	Continuous	One dichotomous, one continuous	Point-Biserial r (see Part 2: Correlation Family)
Categorical	Categorical	Nominal	Chi-Square Test of Independence
Categorical (paired)	Categorical (paired)	Paired nominal	McNemar Test
Continuous	Continuous (controlling 3rd)	Normal, linear, with confound	Partial Correlation

Pearson Product-Moment Correlation

Also known as: *Pearson r, Pearson's correlation coefficient*

What It Does

Pearson r quantifies the strength and direction of the linear relationship between two continuous variables. It ranges from -1 (perfect negative linear relationship) through 0 (no linear relationship) to +1 (perfect positive linear relationship). It asks: "As one variable goes up, does the other tend to go up (or down) in a consistent, straight-line fashion?"

Concrete Example

You examine the relationship between hours of study and exam scores among 100 undergraduates. A Pearson r of .65 would indicate a moderately strong positive relationship: students who study more tend to score higher.

Intuitive Analogy

Imagine plotting everyone's data as dots on a scatterplot. Pearson r measures how closely those dots cluster around a straight line. If they form a tight upward diagonal, r is near +1. If they are a random cloud, r is near 0.

When and Why to Use It

Use when both variables are continuous, approximately normally distributed, and you expect a linear (straight-line) relationship. If the relationship might be curvilinear (U-shaped, for example), Pearson r will underestimate the true association.

Key Assumptions

- Both variables are continuous (interval or ratio).
- The relationship is linear (check with a scatterplot).
- Bivariate normality (both variables are approximately normally distributed).
- No significant outliers (outliers disproportionately affect r).
- Observations are independent.

Formula

$$r = \sum[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2]}$$

r = Pearson correlation coefficient

x_i, y_i = Individual scores on variables X and Y

$\bar{x}, \bar{y}$ = Means of X and Y

Effect Size

r IS the effect size. Benchmarks: |r| = .10 (small), .30 (medium), .50 (large). r^2 = proportion of shared variance.

R Implementation

Package: Base R

```
# Pearson correlation
study_hours <- c(2, 4, 6, 8, 10, 3, 5, 7, 9, 1)
exam_scores <- c(55, 65, 72, 85, 90, 58, 68, 78, 88, 50)
cor.test(study_hours, exam_scores, method = "pearson")

# Correlation matrix for multiple variables
cor(data.frame(study_hours, exam_scores))
```

APA Reporting Template

APA Reporting: *There was a significant strong positive correlation between study hours and exam scores, r(8) = .99, p < .001.*

What If Assumptions Are Violated?

If normality is violated or variables are ordinal, use Spearman's rho. If outliers are present, consider a robust correlation (e.g., percentage-bend correlation from the WRS2 package).

Spearman Rank-Order Correlation

Also known as: *Spearman's rho (ρ), Spearman's rank correlation*

What It Does

Spearman's rho is the non-parametric version of Pearson r. It converts both variables to ranks, then calculates the Pearson correlation on those ranks. This captures any monotonic relationship (consistently increasing or decreasing), not just linear ones.

Concrete Example

You examine the relationship between socioeconomic status (ranked) and self-reported health (ranked on a scale). Spearman's rho tests whether higher SES tends to go with higher health, regardless of whether the relationship is a perfect straight line.

Intuitive Analogy

Instead of measuring exact distances between dots and a line, you are asking: Do people who rank high on one variable also rank high on the other? It is like comparing class rank in math to class rank in science — it does not matter what the actual grades are, just the ordering.

When and Why to Use It

Use when variables are ordinal, when data are non-normal, when there are outliers, or when the relationship is monotonic but not necessarily linear.

Key Assumptions

- Both variables are at least ordinal.
- The relationship is monotonic (consistently increasing or decreasing).
- Observations are independent.

Formula

$$r_s = 1 - [6\sum d^2_i / n(n^2 - 1)]$$

r_s = Spearman's rank correlation coefficient

d_i = Difference between ranks of observation i on the two variables

n = Number of observations

Note: This shortcut formula assumes no tied ranks. When ties are present (common with Likert-type data), R's cor.test() uses the full Pearson-on-ranks algorithm, which handles ties correctly. Hand-calculated values using this formula may not match software output when ties exist.

Effect Size

r_s IS the effect size. Same benchmarks as Pearson r: .10, .30, .50.

R Implementation

Package: Base R

```
# Spearman correlation
ses_rank <- c(1, 3, 5, 2, 7, 4, 6, 8, 9, 10)
health_rank <- c(2, 4, 6, 1, 8, 3, 5, 7, 10, 9)
cor.test(ses_rank, health_rank, method = "spearman")
```

APA Reporting Template

APA Reporting: *There was a significant strong positive monotonic relationship between SES and health, r_s(8) = .90, p < .001.*

What If Assumptions Are Violated?

Very robust. For very small samples, exact p-values may be computed. If you want to handle ties more carefully, use Kendall's tau.

Chi-Square Test of Independence

Also known as: *χ^2 test of independence, Pearson chi-square*

What It Does

This test determines whether two categorical variables are associated. It compares observed frequencies in a contingency table to the frequencies you would expect if the variables were completely independent. If the observed and expected frequencies differ substantially, the variables are related.

Concrete Example

You survey 200 people on political affiliation (Democrat, Republican, Independent) and support for a policy (Yes, No). The chi-square test tells you whether support for the policy depends on political affiliation.

Intuitive Analogy

Imagine surveying everyone in a large office about two things: whether they prefer coffee or tea, and whether they work in the morning shift or the evening shift. If coffee-and-tea preferences are distributed the same way regardless of shift, the two variables are independent. But if morning-shift workers overwhelmingly prefer coffee while evening-shift workers lean toward tea, the distribution departs from what you would expect by chance — the variables are associated. The chi-square test formalizes that comparison between what you observed and what you would expect if the variables were truly unrelated.

When and Why to Use It

Use when both variables are categorical (nominal or ordinal treated as nominal) and you want to test whether they are independent. Requires a sufficiently large sample — generally, all expected cell frequencies should be 5 or more.

Key Assumptions

- Both variables are categorical.
- Observations are independent (each case contributes to only one cell).
- Expected cell frequencies are at least 5 in 80% of cells (no cell below 1).
- For 2×2 tables with small samples, use Fisher's exact test instead.

Formula

$$\chi^2 = \sum [(O - E)^2 / E]$$

χ^2 = Chi-square test statistic

O = Observed frequency in each cell

E = Expected frequency = (row total × column total) / grand total

df = (rows - 1)(columns - 1)

Effect Size

Cramér's V = $\sqrt{(\chi^2 / (N \times \min(r-1, c-1)))}$. Benchmarks depend on df: for 2×2, V = .10 (small), .30 (medium), .50 (large). Phi coefficient for 2×2 tables.

R Implementation

Package: Base R; vcd package for Cramér's V

```
# Chi-square test of independence
data <- matrix(c(40, 30, 20, 25, 35, 50), nrow = 2,
               dimnames = list(c("Support","Oppose"),
                               c("Dem","Rep","Ind")))
chisq.test(data)

# Effect size: Cramer's V
library(vcd)
assocstats(data)
```

APA Reporting Template

APA Reporting: *A chi-square test of independence indicated a significant association between political affiliation and policy support, $\chi^2(2, N = 200) = 12.34$, $p = .002$, $V = .25$.*

What If Assumptions Are Violated?

If expected frequencies are too low, use Fisher's exact test: fisher.test(data). For 2×2 tables, consider Yates' continuity correction (applied by default in R).

McNemar Test

Also known as: *McNemar's chi-square test*

What It Does

McNemar's test is for paired or repeated categorical data. It tests whether the proportion of cases that change in one direction is different from the proportion that change in the other direction. It focuses on the discordant pairs — cases where the outcome changed.

Concrete Example

Before a campaign, 60% of voters support a policy. After the campaign, you survey the same voters again. McNemar's test asks whether significantly more people switched from oppose-to-support than from support-to-oppose.

Intuitive Analogy

Imagine a before-and-after taste test where people choose Coke or Pepsi. McNemar's test ignores those who stayed loyal and focuses on the switchers: Did more people switch from Coke to Pepsi than the reverse?

When and Why to Use It

Use when you have paired binary data — the same participants measured at two time points (or two matched conditions) on a dichotomous outcome.

Key Assumptions

- Paired dichotomous data.
- The sum of discordant pairs should be at least 10 for the chi-square approximation.
- Observations are independent across pairs.

Formula

$$\chi^2 = (b - c)^2 / (b + c)$$

b = Number who changed from No to Yes

c = Number who changed from Yes to No

$b + c$ = Total number of discordant pairs

Effect Size

Odds ratio of discordant pairs = b/c. Or report the exact proportions of change.

R Implementation

Package: Base R

```
# McNemar test
# Rows = Before, Columns = After
data <- matrix(c(40, 15, 5, 40), nrow = 2,
   dimnames = list("Before" = c("Support","Oppose"),
   "After" = c("Support","Oppose")))
mcnemar.test(data, correct = FALSE)
```

APA Reporting Template

APA Reporting: *McNemar's test (without continuity correction) indicated a significant change in policy support after the campaign, $\chi^2(1) = 5.00$, $p = .025$, with significantly more people switching to support ($n = 15$) than switching away ($n = 5$).*

What If Assumptions Are Violated?

For small samples (discordant pairs < 10), use the exact binomial test: binom.test(b, b + c, p = 0.5), where b and c are the discordant cell counts. For moderate samples, applying the continuity correction via mcnemar.test(data, correct = TRUE) (the default in R) provides a more conservative chi-square approximation.

Partial Correlation

Also known as: *Partial r*

What It Does

A partial correlation measures the linear relationship between two continuous variables after statistically removing ("partialing out") the effect of one or more other variables. It reveals the unique relationship between X and Y that is not explained by the control variable(s).

Concrete Example

You find a correlation between ice cream sales and drowning rates. But both are driven by temperature. A partial correlation between ice cream sales and drowning rates, controlling for temperature, reveals whether any direct association remains once the shared influence of temperature is removed.

Intuitive Analogy

Imagine two people are on a tandem bicycle (moving together because of a shared mechanism). A partial correlation removes the bicycle — it asks whether the two riders would still move in sync if they were walking independently.

When and Why to Use It

Use when you want to examine the relationship between two variables while controlling for a confound. It is simpler than regression when you just need a single controlled association.

Key Assumptions

- All variables are continuous.
- Linear relationships among all variables.
- Normality of all variables.
- No significant outliers.

Formula

$$r_xy.z = (r_xy - r_xz \cdot r_yz) / \sqrt{[(1 - r^2_xz)(1 - r^2_yz)]}$$

r_xy.z = Partial correlation of X and Y, controlling for Z

r_xy = Pearson correlation between X and Y

r_xz, r_yz = Correlations of X and Y with the control variable Z

Effect Size

The partial r itself is the effect size. Same benchmarks as Pearson r.

R Implementation

Package: ppcor package

```
library(ppcor)
df <- data.frame(
  icecream = c(10,15,20,25,30,12,18,22,28,35),
  drowning = c(2, 3, 5, 6, 8, 2, 4, 5, 7, 9),
  temp     = c(70,75,80,85,90,72,78,82,88,95))
pcor.test(df$icecream, df$drowning, df$temp)
```

APA Reporting Template

APA Reporting: *After controlling for temperature, the partial correlation between ice cream sales and drowning rates was no longer significant, r(7) = .12, p = .42.*

What If Assumptions Are Violated?

Non-linear relationships will not be fully captured. Consider semi-partial (part) correlations if you only want to remove the control variable's influence from one of the two variables.

Branch 3: Predicting Outcomes (Regression)

You want to predict a dependent variable from one or more independent variables, or understand how much variance predictors explain. The key question: What type is your DV?

DV Type	Predictors	Model
Continuous	1 continuous	Simple Linear Regression
Continuous	2+ continuous/categorical	Multiple Linear Regression
Continuous	2+ (entered in blocks)	Hierarchical Regression
Binary (0/1)	1+	Binary Logistic Regression
Ordinal (ordered categories)	1+	Ordinal Logistic Regression (see Part 2: Regression Family)
Nominal (3+ unordered categories)	1+	Multinomial Logistic Regression (see Part 2: Regression Family)
Count (0, 1, 2, ...)	1+	Poisson Regression

Simple Linear Regression

Also known as: *Bivariate regression, OLS regression (with one predictor)*

What It Does

Simple linear regression models the relationship between one predictor (X) and one continuous outcome (Y) as a straight line. It estimates the intercept (where the line crosses zero) and the slope (how much Y changes for a one-unit increase in X). It also tests whether the predictor significantly improves prediction over simply using the mean.

Concrete Example

You predict exam scores from study hours. The regression equation might be: Score = 45 + 5.2(Hours). This means a student who studies zero hours would be predicted to score 45, and each additional hour of study is associated with a 5.2-point increase.

Intuitive Analogy

Regression draws the best-fitting straight line through a scatterplot — the line that minimizes the total squared distance between each point and the line. It is like finding the ideal trend line that captures the overall direction of the data.

When and Why to Use It

Use when you want to predict a continuous DV from a single continuous (or dichotomous) predictor and understand the nature of their linear relationship.

Key Assumptions

- The relationship between X and Y is linear.
- Independence of observations.
- Homoscedasticity: the spread of residuals is constant across levels of X.
- Normality of residuals (not of the raw variables).
- No significant outliers or influential cases.

Formula

$$\hat{Y} = b_0 + b_1 X$$

$\hat{Y}$ = Predicted value of Y

b_0 = Intercept (value of Y when X = 0)

b_1 = Slope (change in Y per one-unit change in X)

$b_1 = r(s_y / s_x)$ = Slope formula using correlation and standard deviations

Effect Size

R^2 = proportion of variance in Y explained by X. Benchmarks: .01 (small), .09 (medium), .25 (large). Adjusted R^2 penalizes for number of predictors.

R Implementation

Package: Base R

```
# Simple linear regression
hours <- c(2, 4, 6, 8, 10, 3, 5, 7, 9, 1)
scores <- c(55, 65, 72, 85, 90, 58, 68, 78, 88, 50)
model <- lm(scores ~ hours)
summary(model)

# Confidence intervals for coefficients
confint(model)
```

APA Reporting Template

APA Reporting: *A simple linear regression revealed that study hours significantly predicted exam scores, b = 4.68, SE = 0.25, t(8) = 18.72, p < .001. Study hours explained 97.8% of the variance in exam scores, R^2 = .978, F(1, 8) = 350.4, p < .001.*

What If Assumptions Are Violated?

If residuals are non-normal, consider a transformation or robust regression. If heteroscedasticity is present, use robust standard errors: coeftest(model, vcov = vcovHC(model)) from the sandwich and lmtest packages.

Multiple Linear Regression

Also known as: *Multiple regression, OLS regression*

What It Does

Multiple regression predicts a continuous DV from two or more predictors simultaneously. Each predictor's coefficient represents its unique contribution, holding all other predictors constant. This allows you to determine which predictors matter most and how they work together.

Concrete Example

You predict job performance from years of experience, conscientiousness, and education level. Multiple regression tells you how much each predictor uniquely contributes after accounting for the others.

Intuitive Analogy

If simple regression is adjusting one knob on a sound mixer, multiple regression adjusts all the knobs simultaneously to find the best combination that produces the clearest signal (prediction).

When and Why to Use It

Use when you have a continuous DV and two or more predictors (continuous or categorical/dummy-coded). This is the workhorse of social science prediction.

Key Assumptions

- Linear relationship between each predictor and the DV.
- Independence of observations.
- Homoscedasticity of residuals.
- Normality of residuals.
- No multicollinearity: predictors should not be too highly correlated with each other (VIF < 5–10).
- No significant outliers or influential cases.

Formula

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + ... + b_kX_k$$

b_0 = Intercept

$b_1...b_k$ = Regression coefficients (partial slopes) for each predictor

β *(beta)* = Standardized coefficient — allows comparison of predictor importance

k = Number of predictors

Effect Size

R^2 and Adjusted R^2 for overall model. $f^2 = R^2 / (1 - R^2)$ for Cohen's f-squared. Benchmarks: f^2 = .02 (small), .15 (medium), .35 (large). Also sr^2 (squared semi-partial correlation) for each predictor's unique contribution.

R Implementation

Package: Base R; lm.beta for standardized coefficients; car for VIF

```
# Multiple regression
df <- data.frame(
  performance = c(75,82,68,90,85,72,88,65,92,78),
  experience = c(5,8,3,12,10,4,9,2,15,6),
  conscient = c(4.2,3.8,3.5,4.5,4.0,3.6,4.3,3.2,4.8,3.9),
  education = c(16,18,14,20,16,14,18,12,20,16))
model <- lm(performance ~ experience + conscient + education, data=df)
summary(model)

# Check multicollinearity
library(car)
vif(model)
```

APA Reporting Template

APA Reporting: *A multiple regression analysis revealed that the model significantly predicted job performance, R^2 = .94, F(3, 6) = 32.14, p < .001. Experience (β = .52, p = .003) and conscientiousness (β = .35, p = .021) were significant predictors, while education was not (β = .15, p = .310).*

What If Assumptions Are Violated?

If multicollinearity exists (VIF > 10), consider removing or combining predictors. If residuals are non-normal, try transformations or robust regression. If heteroscedasticity is present, use robust standard errors.

Hierarchical (Sequential) Regression

Also known as: *Sequential regression, blockwise regression*

What It Does

Hierarchical regression enters predictors in theoretically-driven blocks (steps) to determine how much additional variance each block explains beyond the previous blocks. It is not a different statistical technique from multiple regression — it is a strategy for entering predictors in a meaningful order.

Concrete Example

You predict academic achievement in three steps: Block 1 = demographics (SES, gender), Block 2 = prior achievement, Block 3 = motivation. You can ask: After controlling for demographics and prior achievement, does motivation explain additional variance?

Intuitive Analogy

Imagine building a tower layer by layer. Each layer is a block of predictors. You measure how much taller the tower gets with each new layer. If adding the third layer (motivation) does not meaningfully increase the tower's height (R^2), maybe it is not worth the added complexity.

When and Why to Use It

Use when theory dictates a specific order for entering predictors, and you want to test whether later-entered variables add significant predictive power beyond earlier variables. The key output is the ΔR^2 (R-squared change) for each block.

Key Assumptions

- Same as multiple regression.
- The order of entry is theoretically justified (not data-driven).

Formula

$$\Delta R^2 = R^2_full - R^2_reduced; \;\; F_change = (\Delta R^2 / \Delta df) / ((1 - R^2_full) / df_residual)$$

ΔR^2 = Change in R-squared when adding a new block

F_change = Tests whether ΔR^2 is significantly greater than zero

Effect Size

ΔR^2 for each block. f^2 for each block = ΔR^2 / (1 - R^2_full).

R Implementation

Package: Base R (run sequential lm models and compare with anova())

```
# Hierarchical regression
model1 <- lm(achievement ~ ses + gender, data = df)
model2 <- lm(achievement ~ ses + gender + prior_ach, data = df)
model3 <- lm(achievement ~ ses + gender + prior_ach + motivation, data = df)
```

```
# Test each step
anova(model1, model2)  # Does prior_ach add?
anova(model2, model3)  # Does motivation add?
summary(model3)
```

APA Reporting Template

APA Reporting: *In Step 1, demographics explained 15% of the variance, $R^2 = .15$, $F(2, 97) = 8.56$, $p < .001$. In Step 2, prior achievement explained an additional 35% of variance, $\Delta R^2 = .35$, F_change(1, 96) = 62.30, $p < .001$. In Step 3, motivation explained an additional 8%, $\Delta R^2 = .08$, F_change(1, 95) = 18.42, $p < .001$.*

What If Assumptions Are Violated?

Be careful about the order of entry. Data-driven ordering (stepwise) is generally discouraged in favor of theory-driven ordering.

Binary Logistic Regression

Also known as: *Logistic regression, logit model*

What It Does

Logistic regression predicts a binary outcome (yes/no, pass/fail, diagnosed/not) from one or more predictors. Instead of modeling the outcome directly, it models the log-odds (logit) of the outcome occurring. Coefficients represent changes in log-odds, but are typically converted to odds ratios for interpretation.

Concrete Example

You predict whether a student will graduate (yes/no) from GPA, attendance rate, and first-generation status. Logistic regression tells you: for each one-point increase in GPA, how much do the odds of graduating change?

Intuitive Analogy

If linear regression draws a line predicting a number, logistic regression draws an S-shaped curve predicting a probability. It maps any combination of predictors onto a probability between 0 and 1.

When and Why to Use It

Use when your DV is binary (two categories). It is the default regression for categorical outcomes in social science.

Key Assumptions

- Binary DV.
- Independence of observations.
- No multicollinearity among predictors (VIF < 5–10).
- Linear relationship between predictors and the log-odds (logit) of the outcome.
- Large enough sample: at least 10–20 events per predictor variable, where "events" refers to occurrences of the less frequent binary outcome (not total N). For example, in a graduation dataset where 15% of students do not graduate, events = the number of non-graduations.
- No extreme outliers in predictor space.

Formula

$$ln(p / (1-p)) = b_0 + b_1X_1 + b_2X_2 + ... + b_kX_k$$

p = Probability of the outcome (e.g., graduating)

p/(1-p) = Odds of the outcome

ln(p/(1-p)) = Log-odds (logit)

b_1 = Change in log-odds for a one-unit increase in X_1

$e^{\wedge}b_1$ = Odds ratio — the multiplicative change in odds per unit of X_1

Effect Size

Nagelkerke R^2 (pseudo R-squared). Odds ratios (OR) with 95% CI. OR = 1 means no effect; OR > 1 means increased odds; OR < 1 means decreased odds.

R Implementation

Package: Base R (glm function)

```
# Binary logistic regression
df <- data.frame(
  graduated = c(1,1,1,0,1,0,0,1,1,0),
  gpa = c(3.5,3.8,3.2,2.1,3.6,2.4,2.0,3.4,3.9,2.3),
  attendance = c(90,95,85,60,92,65,55,88,97,58))
model <- glm(graduated ~ gpa + attendance, data=df, family=binomial)
summary(model)

# Odds ratios with CI
exp(cbind(OR = coef(model), confint(model)))
```

APA Reporting Template

APA Reporting: *A binary logistic regression indicated that GPA significantly predicted graduation, OR = 4.52, 95% CI [1.23, 16.63], p = .023, while attendance was not a significant predictor, OR = 1.08, 95% CI [0.97, 1.21], p = .162. The model explained 62% of the variance (Nagelkerke R^2 = .62).*

What If Assumptions Are Violated?

If the linearity-of-logit assumption is violated, add interaction or polynomial terms. For complete or quasi-complete separation, consider penalized regression (Firth's method in the logistf package).

Poisson Regression

Also known as: *Log-linear model for counts*

What It Does

Poisson regression models count outcomes (0, 1, 2, 3, ...) as a function of predictors. It assumes the count follows a Poisson distribution, where the mean equals the variance. It uses a log link, so coefficients represent changes in the log of the expected count.

Concrete Example

You predict the number of disciplinary incidents per student per semester from school climate ratings and student demographics.

Intuitive Analogy

Imagine counting the number of cars passing an intersection per hour. Poisson regression helps predict that count based on factors like time of day, weather, and events. It is regression for "how many" questions.

When and Why to Use It

Use when the DV is a count (non-negative integers) and you want to model how predictors influence the frequency of events. If the data show overdispersion (variance > mean), use negative binomial regression instead.

Key Assumptions

- DV is a count (non-negative integers).
- Mean = variance (equidispersion). If violated, consider negative binomial.
- Independence of observations.
- Linear relationship between predictors and the log of the expected count.
- No excess zeros (if so, consider zero-inflated Poisson).

Formula

$$ln(\lambda) = b_0 + b_1X_1 + ... + b_kX_k$$

λ = Expected count (mean of the Poisson distribution)

$ln(\lambda)$ = Log of the expected count

$e^{\wedge}b_1$ = Incidence rate ratio (IRR) — the multiplicative change in expected count per unit of X_1

Effect Size

Pseudo R^2 (McFadden's or Nagelkerke). Incidence rate ratios (IRR) with 95% CI.

R Implementation

Package: Base R (glm with family = poisson)

```
# Poisson regression
df <- data.frame(
  incidents = c(0,1,3,2,5,0,1,4,2,6),
  climate = c(4.5,4.0,3.2,3.8,2.5,4.8,4.2,3.0,3.5,2.0),
  ses = c(3,4,2,3,1,4,3,2,3,1))
model <- glm(incidents ~ climate + ses, data=df, family=poisson)
summary(model)
exp(coef(model))  # IRRs
```

APA Reporting Template

APA Reporting: *A Poisson regression indicated that school climate significantly predicted disciplinary incidents, IRR = 0.42, 95% CI [0.28, 0.63], p < .001, such that each one-unit increase in climate was associated with 58% fewer incidents.*

What If Assumptions Are Violated?

If overdispersed (variance >> mean), use negative binomial regression: glm.nb() from the MASS package. For excess zeros, use zero-inflated Poisson (pscl package).

Branch 4: Examining Structure (Factor Analysis and Dimensionality)

You want to understand the underlying structure of your variables — are your 20 survey items actually measuring 3 latent constructs? Are there natural groupings in your data? The key question: Are you exploring or confirming?

Goal	Approach	Method
Explore unknown structure	Data-driven	EFA (Exploratory Factor Analysis)
Confirm hypothesized structure	Theory-driven	CFA (Confirmatory Factor Analysis)
Reduce variables (no latent theory)	Variance maximization	PCA (Principal Component Analysis)

Exploratory Factor Analysis (EFA)

Also known as: *Factor analysis, common factor analysis*

What It Does

EFA discovers the underlying latent factors that explain the pattern of correlations among observed variables. It tells you how many dimensions your data have and which items load on which factors. Unlike PCA, EFA assumes that shared variance (common factors) is what matters, and separates this from unique variance and error.

Concrete Example

You administer a 30-item personality questionnaire and want to discover how many factors emerge. EFA might reveal that items cluster into five factors corresponding to the Big Five personality traits.

Intuitive Analogy

Imagine you see 30 people moving around a room. At first it looks chaotic, but then you notice they are actually organized into 5 groups, each following a different leader. EFA identifies those leaders (factors) and which people (items) are following each one.

When and Why to Use It

Use when you do not have a strong prior theory about the factor structure and want to discover it from the data. Use CFA (not EFA) if you already have a specific hypothesized structure to test.

Key Assumptions

- Adequate sample size: minimum 5–10 cases per item, ideally 300+ total.
- Sufficient correlations: check with Bartlett's test of sphericity (should be significant) and KMO (Kaiser-Meyer-Olkin; should be > .60).
- Approximate multivariate normality (for maximum likelihood extraction).
- No extreme multicollinearity (no correlations > .90 between items).

Formula

$$X = \Lambda F + \varepsilon$$

$\boldsymbol{X}$ = Vector of observed variables

Λ = Factor loading matrix (pattern of relationships)

F = Latent factors

ε = Unique variance + error for each variable

Effect Size

Factor loadings: |loading| > .30 is minimum for interpretation, > .40 is preferred, > .70 is excellent. Variance explained by each factor.

R Implementation

Package: psych package (fa function); GPArotation for rotations

```
library(psych)
# Determine number of factors
fa.parallel(data, fm = "ml")  # Parallel analysis

# Run EFA with oblimin rotation
efa <- fa(data, nfactors = 3, rotate = "oblimin", fm = "ml")
print(efa, cut = 0.30, sort = TRUE)

# Check KMO and Bartlett's
KMO(data)
cortest.bartlett(cor(data), n = nrow(data))
```

APA Reporting Template

APA Reporting: *An exploratory factor analysis using maximum likelihood extraction and oblimin rotation was conducted on the 30-item questionnaire. Parallel analysis suggested a 3-factor solution, which explained 52.3% of the total variance. All items loaded above .40 on their primary factor.*

What If Assumptions Are Violated?

If KMO is below .60, your data may not be suitable for factor analysis. If multivariate normality is violated, use principal axis factoring (fm = "pa") instead of maximum likelihood.

Confirmatory Factor Analysis (CFA)

Also known as: *Measurement model*

What It Does

CFA tests whether a hypothesized factor structure fits the observed data. Unlike EFA (which is exploratory), you specify exactly which items load on which factors before running the analysis. CFA produces fit indices that tell you how well your theoretical model matches reality.

Concrete Example

You hypothesize that a depression scale has three factors: cognitive, somatic, and affective symptoms. CFA tests whether this 3-factor model adequately reproduces the observed correlations among the items.

Intuitive Analogy

If EFA is like exploring a city without a map, CFA is like having a map and checking whether it matches the actual streets. You already have a theory; CFA tells you if the data support it.

When and Why to Use It

Use when you have a theoretically-derived factor structure to test. Essential for scale validation, measurement invariance testing, and as the measurement model in SEM.

Key Assumptions

- Multivariate normality (or use robust estimators like MLR).
- Sufficient sample size: minimum 200, or 10–20 cases per parameter estimated.
- Correct model specification (the right items load on the right factors).
- No excessive cross-loadings or correlated residuals unless theoretically justified.

Formula

$$\Sigma(\theta) = \Lambda \, \Psi \, \Lambda' + \Theta$$

$\Sigma(\theta)$ = Model-implied covariance matrix

Λ = Factor loading matrix

Ψ = Factor covariance matrix

Θ = Residual (unique) variance matrix

Goal = Minimize the difference between Σ(θ) and the observed S matrix

Effect Size

Fit indices: CFI ≥ .95 (good), RMSEA ≤ .06 (good), SRMR ≤ .08 (good). Factor loadings and R^2 for each item. Omega reliability for each factor.

R Implementation

Package: lavaan package

```
library(lavaan)
model <- '
  cognitive =~ item1 + item2 + item3 + item4
  somatic   =~ item5 + item6 + item7 + item8
  affective =~ item9 + item10 + item11 + item12
'
fit <- cfa(model, data = df, estimator = "MLR")
summary(fit, fit.measures = TRUE, standardized = TRUE)

# Reliability
library(semTools)
compRelSEM(fit)
```

APA Reporting Template

APA Reporting: *A confirmatory factor analysis of the 12-item depression scale supported the hypothesized 3-factor structure, $\chi^2(51) = 82.34$, $p = .004$, CFI = .97, RMSEA = .048 [.025, .069], SRMR = .038. All standardized factor loadings exceeded .50.*

What If Assumptions Are Violated?

If fit is poor, examine modification indices (modindices(fit)) to identify sources of misfit. If multivariate normality is violated, use estimator = "MLR" (robust ML) or estimator = "WLSMV" for ordinal data.

Branch 5: Advanced and Structural Methods

These methods address complex research questions involving mediation, moderation, nested data structures, and full structural equation models.

Mediation Analysis

Also known as: *Indirect effect, mediator model*

What It Does

Mediation tests whether the effect of X on Y operates through an intermediary variable (M). It asks: Does X influence M, and does M in turn influence Y? The indirect effect (a × b) quantifies how much of X's effect on Y passes through M.

Concrete Example

Does social support (X) reduce depression (Y) by increasing self-efficacy (M)? Mediation analysis tests whether social support increases self-efficacy, and self-efficacy in turn reduces depression.

Intuitive Analogy

Think of a chain of dominoes. X pushes M, and M pushes Y. Mediation analysis measures how much of the push from X reaches Y through the M domino, versus how much of X's total effect would remain if you removed M from the chain (the direct effect).

When and Why to Use It

Use when you have a theoretical reason to believe that X affects Y through an intermediary mechanism. The modern approach uses bootstrapping for inference, which is more powerful and accurate than the older Baron and Kenny approach.

Key Assumptions

- Temporal precedence: X should precede M, which should precede Y (ideally).
- Correct model specification (no omitted confounders).
- Linear relationships among X, M, and Y (for standard mediation).
- No measurement error in the mediator (or use SEM-based mediation).
- For bootstrapping: adequate sample size (200+ recommended).

Formula

$$\text{Indirect effect} = a \times b;\ \ \text{Direct effect} = c';\ \ \text{Total effect} = c = c' + ab$$

a = Path from X to M (effect of predictor on mediator)

b = Path from M to Y controlling for X (effect of mediator on outcome)

c' = Direct effect of X on Y controlling for M

c = Total effect of X on Y (without mediator in model)

ab = Indirect effect (the mediated portion)

Effect Size

Proportion mediated = ab / c. Completely standardized indirect effect (ab_cs). Also report the 95% bootstrapped CI for the indirect effect — if it excludes zero, the indirect effect is significant.

R Implementation

Package: lavaan for SEM-based mediation (recommended; shown below). The mediation package (Tingley et al.) provides a formal causal inference framework with sensitivity analysis for unmeasured confounders. processR implements Hayes' PROCESS-style conditional process models. These packages are not interchangeable — choose based on your theoretical framework.

```
library(lavaan)
model <- '
  # Regressions
  mediator ~ a*predictor
  outcome  ~ b*mediator + cp*predictor
  # Indirect and total effects
  indirect := a * b
  total    := cp + a * b
'
fit <- sem(model, data = df, se = "bootstrap", bootstrap = 5000)
summary(fit, ci = TRUE)
parameterEstimates(fit, boot.ci.type = "perc")
```

APA Reporting Template

APA Reporting: *Mediation analysis using bootstrapping (5,000 samples) revealed a significant indirect effect of social support on depression through self-efficacy, ab = -2.34, 95% CI [-3.89, -1.12]. The direct effect was also significant, c' = -1.56, p = .023, suggesting partial mediation.*

What If Assumptions Are Violated?

If the causal ordering is uncertain, mediation results should be interpreted cautiously. Cross-sectional mediation is increasingly criticized. Consider longitudinal designs for stronger causal inference.

Moderation Analysis

Also known as: *Interaction effect, moderator model*

What It Does

Moderation tests whether the relationship between X and Y depends on the level of a third variable (W). It asks: Is the effect of X on Y stronger, weaker, or reversed at different levels of W? Statistically, moderation is tested by including an interaction term (X × W) in the regression model.

Concrete Example

Does the effect of study hours on exam scores depend on test anxiety? Perhaps study hours help more for low-anxiety students than high-anxiety students. If so, anxiety moderates the study-score relationship.

Intuitive Analogy

Moderation is like a volume knob on a stereo. The song (X's effect on Y) is always playing, but the moderator (W) controls how loud it is. At one setting, the effect is strong; at another, it's barely audible.

When and Why to Use It

Use when you theorize that the strength or direction of a relationship changes depending on a third variable. Include the interaction term and both main effects in the model.

Key Assumptions

- Same as multiple regression.
- Adequate sample size to detect interactions (interactions require more power).
- Mean-center continuous predictors before creating the interaction term to reduce multicollinearity.

Formula

$$Y = b_0 + b_1X + b_2W + b_3(X \times W) + \varepsilon$$

b_3 = The interaction coefficient — if significant, moderation exists

$X \times W$ = Product of the predictor and moderator

Simple slopes = The effect of X on Y at specific levels of W (e.g., mean, ±1 SD)

Effect Size

ΔR^2 attributable to the interaction term. f^2 for the interaction. Cohen's f^2 benchmarks: .02, .15, .35.

R Implementation

Package: Base R; interactions package for probing and plotting

```
# Moderation analysis
df$hours_c <- scale(df$hours, scale = FALSE)   # Mean-center
df$anxiety_c <- scale(df$anxiety, scale = FALSE)
model <- lm(score ~ hours_c * anxiety_c, data = df)
summary(model)

# Probe the interaction
library(interactions)
sim_slopes(model, pred = hours_c, modx = anxiety_c)
interact_plot(model, pred = hours_c, modx = anxiety_c)
```

APA Reporting Template

APA Reporting: *A moderation analysis revealed a significant interaction between study hours and test anxiety on exam scores, b = -1.23, SE = 0.45, t(96) = -2.73, p = .008, ΔR^2 = .04. Simple slopes analysis indicated that study hours significantly predicted scores at low anxiety (b = 5.80, p < .001) but not at high anxiety (b = 1.12, p = .34).*

What If Assumptions Are Violated?

Interactions are harder to detect than main effects — you need larger samples. If continuous predictors have restricted range, interactions may be attenuated.

Multilevel Modeling (HLM)

Also known as: *Hierarchical Linear Modeling, mixed-effects models, random-effects models*

What It Does

Multilevel modeling handles data with a nested structure — individuals within groups (students in classrooms, employees in companies, repeated measures within people). Standard

regression assumes independence, but nesting violates this. MLM correctly partitions variance across levels and allows slopes and intercepts to vary across groups.

Concrete Example

You study math achievement in 500 students across 30 schools. Students within the same school are more similar to each other than to students in other schools. MLM accounts for this school-level clustering and lets you model both student-level and school-level predictors simultaneously.

Intuitive Analogy

Standard regression is like measuring everyone's temperature with one thermometer in one room. MLM acknowledges that people are in different rooms (which may have different temperatures) and uses a separate thermometer for each room while still drawing overall conclusions.

When and Why to Use It

Use when your data have a hierarchical/nested structure: repeated measures within individuals, individuals within groups, groups within organizations, etc. Ignoring nesting inflates Type I error rates.

Key Assumptions

- Normally distributed residuals at each level.
- Linear relationships at each level.
- Independence between higher-level units (schools are independent, even if students within them are not).
- Adequate sample at each level: generally 20+ Level 2 units for reliable estimates.

Formula

$$\text{Level 1: } Y_ij = \beta_0 j + \beta_1 j(X_ij) + e_ij; \;\; \text{Level 2: } \beta_0 j = \gamma_{00} + \gamma_{01}(W_j) + u_0 j$$

Y_ij = Outcome for individual i in group j

$\beta_0 j$ = Intercept for group j (varies across groups)

$\beta_1 j$ = Slope for group j (can also vary)

γ_{00} = Grand mean intercept across all groups

$u_0 j$ = Random effect: group j's deviation from the grand mean

e_ij = Individual-level residual

Effect Size

ICC (Intraclass Correlation Coefficient): proportion of total variance at the group level. Marginal R^2 (fixed effects only) and conditional R^2 (fixed + random effects) from the MuMIn package.

R Implementation

Package: lme4 package (lmer function); lmerTest for p-values

```
library(lme4)
library(lmerTest)

# Random intercept model
model <- lmer(math_score ~ ses + school_quality + (1 | school_id),
              data = df)
```

```
summary(model)

# ICC
library(performance)
icc(model)

# R-squared
library(MuMIn)
r.squaredGLMM(model)
```

APA Reporting Template

APA Reporting: *A multilevel model with students nested in schools revealed that student SES significantly predicted math achievement, γ = 3.45, SE = 0.82, t = 4.21, p < .001, as did school quality, γ = 5.12, SE = 1.56, t = 3.28, p = .003. The ICC was .22, indicating that 22% of variance in math scores was between schools.*

What If Assumptions Are Violated?

If Level 2 sample is small (< 20 groups), estimates of random effects may be unreliable. Consider Bayesian approaches with the brms package. If residuals are non-normal, consider generalized linear mixed models (glmer).

Structural Equation Modeling (SEM)

Also known as: *Covariance structure modeling, LISREL-type models*

What It Does

SEM is a comprehensive framework that combines factor analysis (measurement models) with path analysis (structural models) to test complex theoretical models. It simultaneously estimates relationships among latent variables (unobserved constructs measured by indicators) and handles measurement error explicitly.

Concrete Example

You model the relationship between "attachment security" (latent, measured by 5 items) and "relationship satisfaction" (latent, measured by 4 items), with "communication quality" (latent, 3 items) as a mediator. SEM tests the full theoretical model at once.

Intuitive Analogy

If regression is like testing one wire in a circuit, SEM tests the entire circuit board at once — all the connections, all the components, and whether the overall design works as theorized.

When and Why to Use It

Use when you have latent variables (measured by multiple indicators), multiple interconnected pathways, and a strong theoretical model to test. SEM is the gold standard for testing complex theoretical models in social science.

Key Assumptions

- Multivariate normality (or use robust estimators).
- Sufficiently large sample (200+ minimum, 10–20 cases per estimated parameter).
- Correct model specification.
- Appropriate indicators for each latent variable.

Formula

$$Measurement: X = \Lambda\xi + \delta;\ \ Structural: \eta = B\eta + \Gamma\xi + \zeta$$

ξ = Exogenous (predictor) latent variables

η = Endogenous (outcome) latent variables

Λ = Factor loadings (measurement model)

B = Paths among endogenous latent variables

Γ = Paths from exogenous to endogenous latent variables

ζ = Structural residuals (disturbances)

Effect Size

Model fit: CFI ≥ .95, RMSEA ≤ .06, SRMR ≤ .08. Standardized path coefficients (β) for individual paths. R^2 for each endogenous variable.

R Implementation

Package: lavaan package

```
library(lavaan)
model <- '
  # Measurement model
  attachment  =~ att1 + att2 + att3 + att4 + att5
  communication =~ com1 + com2 + com3
  satisfaction =~ sat1 + sat2 + sat3 + sat4
  # Structural model
  communication ~ a*attachment
  satisfaction  ~ b*communication + cp*attachment
  # Indirect effect
  indirect := a*b
'
fit <- sem(model, data = df, estimator = "MLR")
summary(fit, fit.measures = TRUE, standardized = TRUE)
```

APA Reporting Template

APA Reporting: *The structural equation model demonstrated good fit, $\chi^2(51) = 78.34$, $p = .008$, CFI = .97, RMSEA = .045 [.020, .067], SRMR = .041. Attachment security significantly predicted communication quality ($\beta = .54$, $p < .001$), which in turn predicted relationship satisfaction ($\beta = .62$, $p < .001$).*

What If Assumptions Are Violated?

If fit is poor, do NOT blindly modify the model to improve fit — modifications should be theoretically justified. For non-normal data, use MLR estimator. For ordinal indicators, use WLSMV.

PART 2: Categorical Quick-Reference

This section groups all tests by family for quick lookup. Each entry provides the test name, a one-sentence purpose, key assumptions, and the R function. Cross-references point to the detailed entry in Part 1 where applicable.

T-Test Family

One-Sample t-Test

Purpose: Tests whether a sample mean differs from a known value.

- Continuous DV
- Normal distribution (or n > 30)
- Independent observations

R: `t.test(x, mu = value)`

See Part 1, Branch 1: Comparing Groups

Independent-Samples t-Test

Purpose: Compares means of two independent groups.

- Continuous DV
- Independence between groups
- Normality in each group
- Equal variances (use Welch's if not)

R: `t.test(x, y) or t.test(x, y, var.equal = TRUE)`

See Part 1, Branch 1: Comparing Groups

Paired-Samples t-Test

Purpose: Compares means from two measurements on the same participants.

- Continuous difference scores
- Normal distribution of differences
- Independent pairs

R: `t.test(x, y, paired = TRUE)`

See Part 1, Branch 1: Comparing Groups

ANOVA Family

One-Way ANOVA

Purpose: Tests whether means differ across 3+ independent groups.

- Continuous DV
- Independence
- Normality per group
- Homogeneity of variance

R: `aov(dv ~ group); summary()`

See Part 1, Branch 1

Factorial ANOVA

Purpose: Tests main effects and interactions of 2+ IVs on a DV.

- Same as one-way ANOVA
- Balanced designs preferred
- Use Type III SS for unbalanced

R: `options(contrasts = c("contr.sum", "contr.poly")); aov(dv ~ iv1 * iv2); Anova(model, type=3)`

See Part 1, Branch 1

Repeated-Measures ANOVA

Purpose: Tests whether means change across 3+ conditions for the same participants.

- Continuous DV
- Sphericity (Mauchly's test)
- Normality at each timepoint

R: `ezANOVA(data, dv, wid, within) from ez package`

See Part 1, Branch 1

Mixed ANOVA

Purpose: Combines between-subjects and within-subjects factors.

- Continuous DV
- Sphericity for within factor
- Homogeneity for between factor

R: `ezANOVA(data, dv, wid, within, between)`

See Part 1, Branch 1

ANCOVA

Purpose: Compares group means while controlling for a continuous covariate.

- ANOVA assumptions plus
- Homogeneity of regression slopes
- Linear covariate-DV relationship

R: `options(contrasts = c("contr.sum", "contr.poly")); aov(dv ~ covariate + group); Anova(model, type=3)`

See Part 1, Branch 1

MANOVA

Purpose: Tests group differences on multiple DVs simultaneously.

- Multiple continuous DVs
- Multivariate normality
- Homogeneity of covariance matrices

R: `manova(cbind(dv1,dv2) ~ group); summary(, test='Pillai')`

See Part 1, Branch 1

MANCOVA

Purpose: MANOVA with continuous covariates.

- MANOVA assumptions plus ANCOVA assumptions

R: `manova(cbind(dv1,dv2) ~ covariate + group)`

Combination of MANOVA and ANCOVA approaches

Correlation Family

Pearson r

Purpose: Measures linear association between two continuous variables.

- Both continuous
- Linear relationship
- Bivariate normality
- No extreme outliers

R: `cor.test(x, y, method = 'pearson')`

See Part 1, Branch 2

Spearman rho

Purpose: Measures monotonic association using ranks.

- At least ordinal
- Monotonic relationship

R: `cor.test(x, y, method = 'spearman')`

See Part 1, Branch 2

Kendall tau

Purpose: Measures ordinal association; more robust than Spearman for small samples or many ties.

- At least ordinal
- Robust to ties and small n

R: `cor.test(x, y, method = 'kendall')`

Useful when Spearman is unreliable due to ties

Point-Biserial r

Purpose: Correlates a dichotomous variable with a continuous variable.

- One dichotomous, one continuous
- Normality in each group

R: `cor.test(x, y) where x is coded 0/1`

Mathematically equivalent to Pearson r with a dichotomy

Partial Correlation

Purpose: Correlation between two variables controlling for a third.

- All continuous
- Linear relationships

R: `pcor.test() from ppcor package`

See Part 1, Branch 2

Semi-Partial (Part) Correlation

Purpose: Removes control variable's influence from only one variable.

- All continuous
- Linear relationships

R: `spcor.test() from ppcor package`

Related to unique R^2 contribution in regression

Regression Family

Simple Linear Regression

Purpose: Predicts continuous DV from one predictor.

- Linear relationship
- Normal residuals
- Homoscedasticity

R: `lm(y ~ x)`

See Part 1, Branch 3

Multiple Linear Regression

Purpose: Predicts continuous DV from 2+ predictors.

- Same as simple + no multicollinearity (VIF < 10)

R: `lm(y ~ x1 + x2 + x3)`

See Part 1, Branch 3

Hierarchical Regression

Purpose: Enters predictors in theory-driven blocks to test incremental validity.

- Same as multiple regression
- Theoretically justified block order

R: `anova(model1, model2) to test` ΔR^2

See Part 1, Branch 3

Binary Logistic Regression

Purpose: Predicts binary outcome from predictors.

- Binary DV
- Independence
- No multicollinearity
- Linearity of logit

R: `glm(y ~ x, family = binomial)`

See Part 1, Branch 3

Ordinal Logistic Regression

Purpose: Predicts ordered categorical outcome.

- Ordinal DV
- Proportional odds assumption

R: `polr(y ~ x, data, Hess=TRUE) from MASS package`

Use Brant test for proportional odds assumption

Multinomial Logistic Regression

Purpose: Predicts unordered categorical outcome with 3+ levels.

- Nominal DV with 3+ categories
- Independence
- No multicollinearity

R: `multinom(y ~ x, data) from nnet package`

Each category compared against a reference

Poisson Regression

Purpose: Models count outcomes.

- Count DV
- Mean = Variance (equidispersion)
- Log-linear relationship

R: `glm(y ~ x, family = poisson)`

See Part 1, Branch 3

Non-Parametric Alternatives

Parametric Test	Non-Parametric Alternative	R Function
One-sample t-test	Wilcoxon signed-rank (vs. value)	wilcox.test(x, mu = value)
Independent t-test	Mann-Whitney U	wilcox.test(x, y)
Paired t-test	Wilcoxon signed-rank	wilcox.test(x, y, paired = TRUE)
One-way ANOVA	Kruskal-Wallis H	kruskal.test(y ~ group)
RM-ANOVA	Friedman test	friedman.test(matrix)
Pearson r	Spearman rho	cor.test(x, y, method='spearman')

Reliability and Validity Tests

Cronbach's Alpha (α)

Purpose: Measures internal consistency (how closely related items are as a group).

- Items measure same construct
- Tau-equivalent items (equal factor loadings) for accurate alpha

R: `alpha(data) from psych package`

Alpha > .70 acceptable; .80+ good. Consider McDonald's omega as superior alternative.

McDonald's Omega (ω)

Purpose: A better measure of composite reliability that does not assume equal factor loadings.

- Items load on a common factor

R: `omega(data) from psych; or compRelSEM() from semTools`

Preferred over alpha in modern psychometrics

Intraclass Correlation Coefficient (ICC)

Purpose: Measures agreement or consistency between raters or repeated measures.

- Continuous ratings
- Normally distributed

R: `ICC(data) from psych; or icc() from irr`

Multiple forms: ICC(1), ICC(2,1), ICC(2,k), ICC(3,1), ICC(3,k)

Cohen's Kappa (κ)

Purpose: Measures inter-rater agreement for categorical ratings, corrected for chance.

- Categorical ratings
- Two raters
- Same categories used by both

R: `kappa2(data) from irr package`

κ: .01-.20 slight, .21-.40 fair, .41-.60 moderate, .61-.80 substantial, .81-1 excellent

PART 3: Supplementary Material

Assumption-Checking Guide

Before running any parametric test, check its assumptions. Here are the most common checks with R code.

Normality

The Shapiro-Wilk test and Q-Q plots are your primary tools. The Shapiro-Wilk tests the null hypothesis that data come from a normal distribution — a significant result ($p < .05$) means the data are non-normal. However, with large samples, even trivial deviations trigger significance, so always pair the test with a Q-Q plot for visual inspection.

```
# Shapiro-Wilk test
shapiro.test(x)

# Q-Q plot
qqnorm(x); qqline(x, col = "red")

# For multivariate normality (Mardia's test)
library(MVN)
mvn(data, mvnTest = "mardia")
```

Homogeneity of Variance

Levene's test checks whether the variances are equal across groups. A significant result means unequal variances, suggesting you should use Welch's t-test or Welch's ANOVA instead of the standard versions.

```
# Levene's test
library(car)
leveneTest(dv ~ group, data = df)

# Bartlett's test (more sensitive, assumes normality)
bartlett.test(dv ~ group, data = df)
```

Multicollinearity

Variance Inflation Factor (VIF) measures how much the variance of a regression coefficient is inflated due to correlations with other predictors. VIF > 5 is concerning; VIF > 10 is severe.

```
# VIF in regression
library(car)
model <- lm(y ~ x1 + x2 + x3, data = df)
vif(model)

# Correlation matrix to inspect bivariate collinearity
cor(df[, c("x1", "x2", "x3")])
```

Outlier Detection

Outliers can distort results. Use multiple methods to identify them.

```
# Z-scores (|z| > 3.29 = potential outlier)
z_scores <- scale(x)
which(abs(z_scores) > 3.29)

# Mahalanobis distance (multivariate outliers)
```

```
md <- mahalanobis(data, colMeans(data), cov(data))
which(md > qchisq(0.999, df = ncol(data)))

# Cook's distance (influential cases in regression)
model <- lm(y ~ x, data = df)
cooks <- cooks.distance(model)
which(cooks > 4 / nrow(df))
```

Sphericity (for Repeated-Measures Designs)

Mauchly's test checks the sphericity assumption. If violated, apply Greenhouse-Geisser or Huynh-Feldt corrections. The ezANOVA function in R automatically provides Mauchly's test and corrections.

```
# Mauchly's test (built into ezANOVA output)
library(ez)
result <- ezANOVA(data=df, dv=score, wid=id, within=time)
result$Mauchly  # Mauchly's test
result$Sphericity  # Corrected results
```

Effect Size Interpretation Table

Effect sizes quantify the magnitude of a finding independent of sample size. Always report them alongside p-values.

Effect Size	Small	Medium	Large	Used With
Cohen's d	0.20	0.50	0.80	t-tests
Eta-squared (η^2)	0.01	0.06	0.14	ANOVA
Partial η^2	0.01	0.06	0.14	Factorial/RM ANOVA
Omega-squared (ω^2)	0.01	0.06	0.14	ANOVA (less biased)
Cohen's f	0.10	0.25	0.40	ANOVA (related to η^2)
r (correlation)	0.10	0.30	0.50	Correlations, non-parametric
R^2 §	0.01	0.09	0.25	Regression
Cohen's f^2	0.02	0.15	0.35	Regression (R^2-based)
Odds Ratio ‡	1.5	2.5	4.3	Logistic regression
Cramér's V (df=1)*	0.10	0.30	0.50	Chi-square
Cramér's V (df=2)*	0.07	0.21	0.35	Chi-square
Cohen's κ †	0.40	0.60	0.80	Inter-rater agreement

Important: These benchmarks are general guidelines from Cohen (1988). Always interpret effect sizes in the context of your specific research domain. A "small" effect can be practically significant in some contexts.

† Cohen's κ is conventionally interpreted using the Landis & Koch (1977) scale rather than Cohen's small/medium/large framework: κ = .01–.20 (slight), .21–.40 (fair), .41–.60 (moderate), .61–.80 (substantial), .81–1.00 (almost perfect). The small/medium/large values in this table are approximate equivalents provided for reference, but the Landis & Koch scale is the dominant interpretive framework for κ in social science.

‡ Odds ratio benchmarks vary across sources. The values here (1.5, 2.5, 4.3) are commonly cited in textbooks. An alternative set derived from converting Cohen's d thresholds (Chen, Cohen, & Chen, 2010) yields 1.68 (small), 3.47 (medium), and 6.71 (large). Researchers should consider domain-specific conventions when interpreting odds ratios.

* For Cramér's V, df = min(r − 1, c − 1), where r = number of rows and c = number of columns in the contingency table. This is not the same as the chi-square test's degrees of freedom, which equals (r − 1)(c − 1). For example, a 2 × 3 table has a chi-square df of 2, but Cramér's V df = min(1, 2) = 1, so the df=1 benchmarks apply.

§ The R^2 benchmarks of .01, .09, and .25 derive from squaring Cohen's (1988) r benchmarks ($.10^2$, $.30^2$, $.50^2$) and are appropriate when R^2 represents a bivariate association (simple regression or squared correlation). In multiple regression contexts, benchmarks derived from Cohen's f^2 thresholds (.02, .15, .35) via the formula $R^2 = f^2/(1 + f^2)$ yield slightly different values: .02 (small), .13 (medium), .26 (large). Researchers should apply the framework appropriate to their analysis: the r-derived benchmarks for bivariate models and the f^2-derived benchmarks for multiple regression.

When Things Go Wrong: Troubleshooting Guide

Assumptions will sometimes be violated. Here is what to do.

Problem	What to Do	R Solution
Non-normal data	Use non-parametric test, or bootstrap, or transform	wilcox.test(); boot package; log(x)
Unequal variances	Use Welch's version of t-test or ANOVA	t.test() (default); oneway.test()
Sphericity violated	Apply Greenhouse-Geisser or Huynh-Feldt correction	ezANOVA applies automatically
Multicollinearity	Remove or combine correlated predictors; use ridge/LASSO	vif(); glmnet package
Outliers	Winsorize, trim, or use robust methods	WRS2 package; robustbase package
Small sample (n < 30)	Use exact tests, bootstrapping, or non-parametric tests	exact = TRUE in tests; boot()
Heteroscedasticity	Use robust standard errors	coeftest(m, vcov=vcovHC(m)) from sandwich
Non-linear relationship	Add polynomial terms, use GAMs, or transform	poly(x,2); gam() from mgcv
Missing data	Multiple imputation or full information ML	mice package; fiml in lavaan
Expected freq < 5 (chi-sq)	Use Fisher's exact test	fisher.test()

Sample Size and Power Analysis

Statistical power is the probability of detecting a true effect if one exists. By convention, we aim for power = .80 (80% chance of detecting the effect). Power depends on four interrelated quantities: sample size, effect size, significance level (alpha), and power itself. If you know any three, you can calculate the fourth.

A priori power analysis (before data collection) determines the sample size you need. Post hoc power analysis (after data collection) is generally discouraged.

Power Analysis in R

```
library(pwr)

# t-test: How many per group for medium effect (d=0.5)?
pwr.t.test(d = 0.5, power = 0.80, sig.level = 0.05,
           type = "two.sample")
# Result: n = 64 per group

# Correlation: How many participants for medium r?
pwr.r.test(r = 0.30, power = 0.80, sig.level = 0.05)
# Result: n = 85 (84.07 rounded up)

# ANOVA: 3 groups, medium effect (f=0.25)?
pwr.anova.test(k = 3, f = 0.25, power = 0.80,
               sig.level = 0.05)
# Result: n = 53 per group

# Chi-square: medium effect (w=0.3), 2x3 table?
pwr.chisq.test(w = 0.30, df = 2, power = 0.80,
               sig.level = 0.05)
# Result: N = 107 total

# Regression: medium f2, 3 predictors?
pwr.f2.test(u = 3, f2 = 0.15, power = 0.80,
   sig.level = 0.05)
# Result: v = 77 (total N = v + u + 1 = 81)
```

Rule of Thumb: For common social science effects (d ≈ 0.5, r ≈ .30), you typically need 50–100 participants per group for adequate power. Smaller effects require substantially larger samples. Always do a formal power analysis for your specific design.

Glossary of Key Terms

Alpha (α): The significance threshold, conventionally set at .05. It represents the probability of a Type I error — incorrectly rejecting a true null hypothesis.

ANOVA: Analysis of Variance. An omnibus test comparing means across 3+ groups by analyzing variance ratios.

Assumption: A condition that must be met (or approximately met) for a statistical test to produce valid results.

Beta (β) coefficient: In regression, the standardized slope indicating how many SDs the DV changes per SD change in the predictor.

Bootstrapping: A resampling method that draws thousands of samples (with replacement) from your data to estimate sampling distributions and confidence intervals without relying on distributional assumptions.

Central Limit Theorem: States that the sampling distribution of the mean approaches normality as sample size increases, regardless of the population distribution. Generally kicks in around n = 30.

Confidence Interval (CI): A range of values constructed so that, across repeated samples, a specified percentage (usually 95%) of such intervals would contain the true population parameter. Any single CI either does or does not contain the true value; the percentage refers to the long-run success rate of the procedure. Wider CI = more uncertainty.

Confound: A third variable that influences both the IV and DV, creating a spurious relationship.

Degrees of Freedom (df): The number of values in a calculation that are free to vary. Related to sample size and number of parameters estimated.

Dependent Variable (DV): The outcome variable you are trying to explain or predict.

Effect Size: A standardized measure of the magnitude of a finding, independent of sample size. Essential for practical significance.

Heteroscedasticity: Unequal spread of residuals across levels of a predictor. Violates regression assumptions and biases standard errors.

Homogeneity of Variance: The assumption that different groups have equal variances on the DV.

Homoscedasticity: Equal spread of residuals — the opposite of heteroscedasticity. An assumption of OLS regression.

Independent Variable (IV): The variable hypothesized to cause or predict changes in the DV. The grouping or predictor variable.

Interaction Effect: When the effect of one IV on the DV depends on the level of another IV. Also called moderation.

Latent Variable: An unobserved construct (like 'anxiety' or 'intelligence') that is inferred from observable indicators (survey items, test scores).

Multicollinearity: High correlations among predictor variables in a regression, making it hard to isolate each predictor's unique effect.

Non-Parametric Test: A statistical test that does not assume a specific distribution (like normality). Uses ranks or frequencies instead of raw scores.

Normal Distribution: A symmetric, bell-shaped distribution defined by its mean and standard deviation. Many parametric tests assume normality.

Null Hypothesis (H0): The default assumption that there is no effect, no difference, or no relationship. Statistical tests evaluate evidence against H0.

Odds Ratio (OR): In logistic regression, the multiplicative change in odds of the outcome for a one-unit increase in the predictor. OR = 1 means no change.

Outlier: An observation that is unusually far from other observations. Can distort means, variances, and correlations.

p-value: The probability of obtaining results at least as extreme as the observed data, assuming the null hypothesis is true. Smaller p = stronger evidence against H0.

Power: The probability of correctly detecting a true effect (1 - Type II error rate). Conventionally set at .80.

R-squared (R^2): The proportion of variance in the DV explained by the predictor(s). Ranges from 0 to 1.

Residual: The difference between an observed value and the value predicted by a model. Good models have small, randomly scattered residuals.

Sphericity: The assumption in RM-ANOVA that the variances of differences between all pairs of repeated measures are equal.

Standard Error (SE): The standard deviation of a sampling distribution. Measures the precision of a sample statistic (like a mean or coefficient).

Type I Error: A false positive — rejecting the null hypothesis when it is actually true. Controlled by setting alpha.

Type II Error: A false negative — failing to reject the null hypothesis when there actually is an effect. Related to power.

Variance: The average squared deviation from the mean. A fundamental measure of spread in a distribution.

Final Notes

Statistics is a tool for understanding the world, not an end in itself. The best analysis is one that matches your research question, respects the nature of your data, and produces results that can be interpreted clearly and honestly. When in doubt, consult the decision tree in Part 1, check your assumptions with the tools in Part 3, and report your results with both a p-value and an effect size.

Remember: there is no single "correct" test for every situation. Many analyses are defensible, and the key is transparency — explaining why you chose the test you did and what you did when assumptions were not perfectly met. Good research is honest research.

Good luck with your analyses!